

(a) Retain the full prefill

ONE-TIME SDPA PREFILL



page images

Retained cache C

n context tokens · original positions



(b) Preview the question and score content blocks

qj question tokens
forwarded over C
once

Question → Context Attention

eager preview · per-layer hooks · immediate reduction



Block scores

aggregate layers, heads, queries



TOP-BUDGET RE-SELECTION

b = 5% nominal budget



selected blue blocks define the view; unselected blocks remain in C

$$s_i = \text{Agg}_{l,h,q} \sum_{k \in B_i} a_{q,k}^{l,h}$$

(c) Build a per-query 4D additive mask

$$\mathbf{M}_j \in \mathbb{R}^{B \times H \times Q \times K}$$

one slice per (batch, head)

row t



Allowed keys

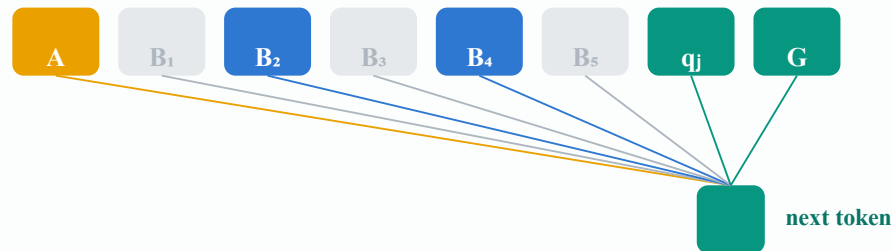
$A \cup S_j \cup q_j \cup G < t$

mask changes;
cache does not

0 = visible to this decode row & $-\infty$ = hidden for this query only

(d) Decode in place under $\mathbf{M}_j$

$$\text{Attn}(\mathbf{Q}, \mathbf{K}, \mathbf{V}; \mathbf{M}_j) = \text{softmax}(\mathbf{Q}\mathbf{K}^T / \sqrt{d} + \mathbf{M}_j) \mathbf{V}$$



NO GATHER · NO EVICTION · NO RE-READ · M-ROPE UNCHANGED

(e) Reversibility: discard the view, retain the memory

qj → mask $\mathbf{M}_j$
selected view

discard $\mathbf{M}_j$ · crop preview
C remains intact

qj+1 → new mask
new view, same cache